

ICL-Traffic Forecasting: Lightweight In-Context Learning for Pretrained Spatio-Temporal Foundation Models : [Applications]

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Abstract

Pretrained spatio-temporal foundation models such as OpenCity give strong zero-shot traffic forecasts, but accuracy drops under cross-city shifts. In NLP and time-series, in-context learning (ICL) adapts predictions at inference time from retrieved demonstrations, yet ICL has not been studied for spatio-temporal foundation models. The standard recipe of feeding demonstrations into the model's context window cannot be applied here: these models are pretrained on fixed-length sequences, and extending them introduces out-of-distribution attention patterns that degrade MAE by 9–28%. We propose a novel traffic prediction application, **ICL-Traffic**, a lightweight ICL module that keeps the base model completely frozen. A 198K-parameter cross-attention aggregator is trained offline in seconds from cached features. At inference time, KNN retrieves the most similar training windows, and the aggregator combines their prediction errors into a per-query correction, with no test-time gradients. On PEMS07M, ICL-Traffic reduces MAE from 4.50 to 4.15 (7.8%) with only ~0.8% extra parameters. On two out-of-distribution cross-city datasets (SZ-DIDI and CD-DIDI), the same module recovers 78–84% of the MAE gain that prediction-head fine-tuning achieves, without modifying a single backbone weight.

Keywords

traffic forecasting, in-context learning, foundation models, spatio-temporal prediction, retrieval-augmented inference

ACM Reference Format:

Parth Ripal Parikh, Mrudani Vishal Hada, Xuesi Zhou, and Ranga Raju Vastavai. 2026. ICL-Traffic Forecasting: Lightweight In-Context Learning for Pretrained Spatio-Temporal Foundation Models : [Applications]. In *Proceedings of November 2026 (ACM SIGSPATIAL '26)*. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

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ACM SIGSPATIAL '26,

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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

Accurate short-term traffic forecasting is foundational to intelligent transportation systems, enabling applications ranging from adaptive signal control and dynamic route guidance to proactive congestion management. As cities grow and traffic patterns become increasingly heterogeneous across regions and time periods, the demand for models that generalize reliably across urban contexts has never been greater. Traditional spatio-temporal models such as STGCN [19] and Graph WaveNet [18] achieve strong supervised performance but require per-dataset training with target-domain labels. Recent work has shifted toward spatio-temporal foundation models (ST-FMs) pretrained on large-scale, heterogeneous traffic data from multiple cities. OpenCity [9] is a representative example: it integrates a Transformer backbone with graph neural networks and achieves zero-shot prediction across cities, data categories, and time horizons without per-dataset retraining. Zero-shot accuracy is compelling, but it degrades when the target city or traffic modality departs substantially from the pretraining distribution. On SZ-DIDI and CD-DIDI, ride-hailing demand datasets from Shenzhen and Chengdu, OpenCity's zero-shot MAE of 4.66 and 6.34 respectively lags full-shot supervised baselines by a substantial margin, exposing the limits of static zero-shot inference under cross-city distribution shift. OpenCity addresses this with 3-epoch prediction-head fine-tuning on target-domain data, but this requires gradient updates. A natural question arises: can demonstration-based adaptation at inference time approximate the gains of fine-tuning for ST-FMs, while keeping the backbone entirely frozen?

In NLP, in-context learning (ICL) enables foundation models to adapt via demonstrations without gradient updates [2]. Chen et al. [3] formalize this as In-Context Tuning (ICT), meta-training a language model to predict outputs from concatenated demonstrations and queries, allowing inference-time generalization. ICT unifies prompting, fine-tuning, and meta-learning, outperforming gradient-based methods like MAML [6] via the language model's pattern-matching inductive bias. Das et al. [4] extend ICL to time-series with In-Context Fine-Tuning (ICF), adapting TimesFM [5] through related time series in its context using separator tokens and cross-example attention. TimesFM-ICF improves by 6.8% on the Monash benchmark and matches per-dataset fine-tuning accuracy, entirely without gradients. Yet output-space, gradient-free ICL, where demonstrations never enter the context and corrections occur in prediction space, remains unexplored for spatio-temporal foundation

models (ST-FMs). Both ICT and ICF modify the base model via meta-training or continued pretraining, but neither transfers to ST-FMs: they assume 1D causal sequences, while ST-FMs operate on structured spatio-temporal graphs with multivariate node features. Crucially, both require architectural retraining, which is prohibitive for ST-FMs given their large-scale pretraining across cities and modalities. Input-space concatenation also fails: OpenCity is pretrained on fixed-length sequences (24 patches = 288 timesteps), so extending the input distorts attention patterns, breaks positional encodings, and disrupts cross-attention, degrading MAE by 9–28% (Section 4.6). Thus, we keep the backbone frozen, process queries and demonstrations independently, and combine them only in the output space.

We propose **ICL-Traffic**, a residual correction framework that enables demonstration-based adaptation for any frozen ST-FM without touching its weights. The central insight is simple: a demonstration’s prediction error estimates the frozen model’s bias on that input, and a learned aggregation network can combine these errors across K retrieved demonstrations into a query-specific correction. Two factors drive improvement: (1) KNN retrieval selects relevant demonstrations from a training-set pool, and (2) a learned aggregator (8K–198K parameters) replaces naive averaging with query-aware weighting. The base model stays fully frozen, and the aggregator trains offline in seconds from cached features. *Our main contributions are:*

- (1) We propose the first output-space, in-context learning application framework for frozen spatio-temporal foundation models that is fully gradient-free and training-free at test time. Unlike FlashST [12] and UniST [20], which require test-time prompt-tuning or parameter updates, our ICL-Traffic application uses demonstration-based corrections entirely in prediction space with no change to backbone weights, enabling lightweight and low-latency inference on resource-constrained devices.
- (2) We systematically study the interaction between demonstration retrieval and correction aggregation. We show that while structured retrieval provides the necessary error signals, it inevitably introduces out-of-distribution noise; our learned aggregator resolves this by acting as a localized spatial-temporal filter to isolate and scale the true systematic bias, showing both axes are jointly necessary to surpass zero-shot performance.
- (3) On PEMS07M, our best configuration reduces MAE by 7.8% ($4.50 \rightarrow 4.15$) with only 198K parameters ($\approx 0.8\%$ of the backbone) and no backbone weight modifications. On out-of-distribution datasets (SZ-DIDI and CD-DIDI), the same module recovers $\approx 80\%$ of the MAE gain of gradient-based prediction-head fine-tuning. Finally, we introduce the first empirical decomposition of adaptation gains for ST-FMs, demonstrating that the remaining $\approx 20\%$ gap defines a structural ‘Representation Adaptation Gap’, quantifying the boundary where post-hoc output-space corrections must yield to representation-level parameter alignment.

2 Related Work

2.1 Spatio-Temporal Foundation Models

Classical spatio-temporal models, including STGCN [19], Graph WaveNet [18], AGCRN [1], and ASTGCN [7], learn dataset-specific parameters and do not transfer across cities. Early ST pretraining approaches such as GPT-ST [11] and STEP [15] demonstrated that self-supervised pretraining on traffic data improves downstream forecasting, laying the groundwork for the full foundation models described below. Recent foundation models address generalization through large-scale pretraining across multiple cities and data types.

OpenCity [9], our base model, integrates Transformer architecture with graph neural networks. It employs instance normalization for distribution shift handling, patch embedding for long-horizon prediction, and temporal-categorical joint attention. Pretrained on heterogeneous multi-city traffic data covering flow, speed, and demand categories, OpenCity achieves zero-shot performance competitive with full-shot baselines across six benchmark datasets and shows promising scaling laws. When zero-shot performance is insufficient, OpenCity provides fast adaptation via 3-epoch fine-tuning of its prediction head on target-domain data. However, this remains gradient-based adaptation, and in-context learning has not been explored for OpenCity or any other ST-FM.

Other ST-FMs explore related adaptation strategies. UniST [20] employs knowledge-guided prompts for few-shot and zero-shot generalization across more than 20 urban spatio-temporal scenarios. FlashST [12] trains a dedicated spatio-temporal prompt network on downstream data and describes the process as in-context learning; unlike ICL-Traffic, however, it requires a separate prompt-training phase that modifies parameters. UrbanGPT [10] uses instruction tuning to generalize across urban tasks via a large language model backbone. Critically, none of these approaches applies corrections in output space with a fully frozen backbone, they all modify parameters either in the prompt network, prediction head, or model weights during adaptation.

Retrieval-augmented approaches have begun to appear in spatio-temporal forecasting. UniFlow [21] introduces Spatio-Temporal Memory Retrieval Augmentation (ST-MRA), which maintains structured memory modules of shared spatio-temporal patterns and retrieves them adaptively during prediction. Unlike ICL-Traffic, ST-MRA is integrated into the model architecture and trained end-to-end, requiring full model training rather than a frozen backbone. Both approaches demonstrate that retrieval of relevant historical context improves spatio-temporal forecasting, validating the core premise of ICL-Traffic, though neither operates under the frozen-backbone constraint that motivates our output-space design.

2.2 In-Context Learning

GPT-3 [2] demonstrated that large language models can adapt to entirely new tasks by conditioning on a small number of labeled demonstrations in the input prompt; a capability termed in-context learning (ICL). Crucially, this requires no weight updates: the model’s pattern-matching ability, developed through pretraining, drives adaptation at inference time. Chen et al. [3] formalize this through In-Context Tuning (ICT): given concatenated instruction, labeled in-context examples, and a target query, the model is meta-trained to predict the target label across diverse NLP tasks.

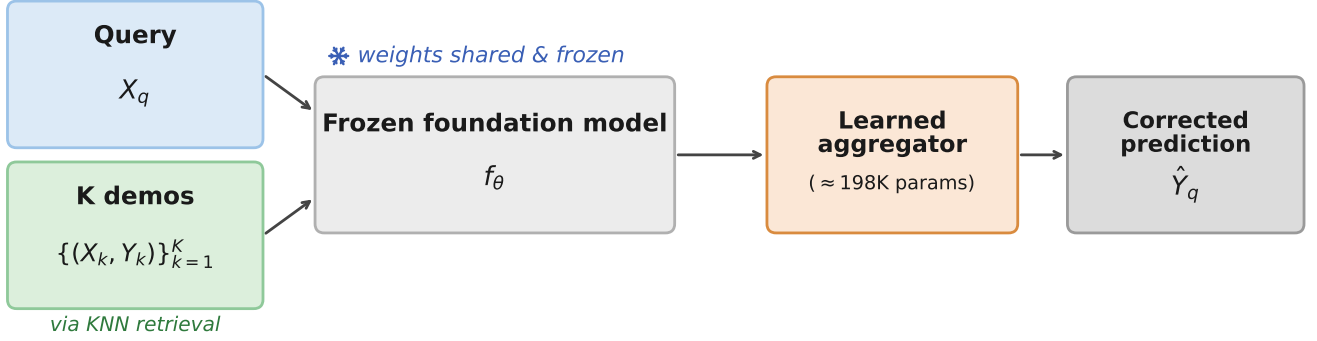


Figure 1: ICL-Traffic pipeline: a query and K KNN-retrieved demonstrations share a frozen foundation model f_θ ; a learned aggregator emits an output-space correction.

ICT outperforms gradient-based meta-learning (MAML [6]) and reduces sensitivity to example ordering and selection. Concurrent work such as MetaICL [13] similarly shows that meta-training a language model on diverse tasks enables rapid in-context adaptation, further validating the ICL paradigm across settings. Das et al. [4] extend ICL to time-series foundation models through In-Context Fine-Tuning (ICF): starting from TimesFM [5], they introduce separator tokens between in-context examples, enable cross-example causal attention, and remove positional encoding. Through continued pretraining, the model learns to borrow patterns from related time series in its context window. TimesFM-ICF achieves a 6.8% improvement over the base TimesFM on the Monash benchmark and matches the accuracy of per-dataset fine-tuning entirely at inference time. These results collectively confirm that retrieved demonstrations improve adaptation in both language and time-series forecasting, yet output-space, gradient-free ICL remains unexplored for spatio-temporal foundation models.

2.3 Relationship to ICT and TimesFM-ICF

ICT and TimesFM-ICF are the closest prior work to ours. All three methods use retrieved demonstrations to adapt a pretrained model at inference time. The critical difference is where the demonstration information acts: ICT and TimesFM-ICF inject demonstrations into the model’s input context, requiring architectural modification and continued training. ICL-Traffic instead applies corrections entirely in output space, keeping the backbone frozen and unchanged. We call this the *input-space* vs. *output-space* distinction, and it is the organizing principle of our design.

This distinction has direct practical consequences for ST-FMs. First, both ICT and TimesFM-ICF feed demonstrations into the model’s context window by concatenating them with the query. We show in Section 4.6 that this fails for OpenCity because its fixed-length pretraining creates out-of-distribution attention patterns, degrading MAE by 9–28%. Second, both methods assume 1-D causal token sequences, whereas ST-FMs process structured spatio-temporal graphs with multi-variate node features, so the concatenation paradigm does not naturally extend to this setting. These observations motivate a fundamentally different design: rather than

forcing demonstrations into the model’s context, we keep the backbone entirely frozen and combine demonstration information only in output space.

We position ICL-Traffic as the output-space surrogate of these methods: it uses the same ingredient (retrieved demonstrations) while keeping the backbone frozen. Our baselines reflect this positioning: (a) the foundation model’s zero-shot inference, (b) random-demonstration residual correction to isolate retrieval quality, and (c) OpenCity’s 3-epoch prediction-head fine-tuning as a gradient-based counterpart at the same training budget.

2.4 Test-Time Adaptation

Test-time adaptation (TTA) updates model parameters at inference time from self-supervised signals. TENT [17] minimises prediction entropy over batch-norm affine parameters; TTT [16] adds a self-supervised auxiliary head that is jointly trained at training time and updated at test time. Both require test-time backward passes and modify the base model. ICL-Traffic takes the opposite approach: the aggregator is trained once offline from cached features, and no gradient computation occurs at test time. The two paradigms are complementary: applying ICL-Traffic first and then running TTA on the residual error is a promising direction for future work.

TTA methods face a fundamental limitation in cross-city traffic scenarios: the self-supervised signals they rely on may not align with the target city’s traffic distribution when the distribution shift is structural rather than covariate. For example, a model pretrained on highway speed data applied to ride-hailing demand in a new city faces a modality shift that entropy minimization cannot resolve, since the model’s output distribution itself is misspecified. ICL-Traffic sidesteps this by using labeled demonstrations from the target domain to directly estimate and correct the model’s bias, rather than relying on unsupervised signals that may be uninformative under large semantic shifts.

3 Method

3.1 Problem Formulation

A traffic network has N sensor nodes. Given a window of historical observations $\mathbf{X} \in \mathbb{R}^{T \times N \times F}$ over T time steps with F features

per node, the task is to predict future values $Y \in \mathbb{R}^{T' \times N}$ over the next T' steps. Let f_θ denote a frozen pretrained ST-FM with parameters θ , producing $\hat{Y} = f_\theta(X)$. We assume a demonstration pool $\mathcal{D} = \{(X_d, Y_d)\}_{d=1}^{|\mathcal{D}|}$ drawn from the target-domain training split, where both inputs and ground-truth futures are available. For a test query X_q , we want to pick K demonstrations from $\mathcal{D}$ and return a prediction $\hat{Y}_q$ that improves on $f_\theta(X_q)$ without touching θ .

3.2 Residual Correction Framework

The core intuition behind ICL-Traffic is that a demonstration from the same traffic domain whose prediction error is known provides a direct signal about the frozen model's systematic bias. By aggregating these error signals from multiple retrieved demonstrations, we can construct a query-specific correction that compensates for the backbone's distribution mismatch without any gradient computation. For each selected demonstration (X_k, Y_k) , we compute the residual correction

$$c_k = Y_k - f_\theta(X_k), \quad (1)$$

which captures the model's bias on input X_k . The corrected prediction for query X_q is

$$\hat{Y}_q = f_\theta(X_q) + \text{Aggregate}\left(\{c_k\}_{k=1}^K, h_q, \{h_k\}_{k=1}^K\right), \quad (2)$$

where h_q, h_k are frozen-encoder features. Because every input is processed through the standard forward path, the input sequence length is never modified. When $K = 0$ the framework falls back to zero-shot prediction. Both demo predictions and encoder features can be cached in a one-time Phase 1 pass, so aggregator training requires no additional forward passes through the base model.

3.3 Demonstration Retrieval

The quality of residual corrections depends critically on similarity between demonstrations and the query. We investigate three retrieval strategies.

Random selection: We sample K demonstrations uniformly from $\mathcal{D}$. Because randomly selected demonstrations have low correlation with the query, their residuals are effectively noise with respect to the query's prediction error. With $K = 3$, random selection yields MAE 6.14, which is 36% worse than zero-shot, confirming that demo quality matters.

Raw-input KNN: We retrieve the K nearest neighbors by L2 distance in the flattened raw input space,

$$\mathcal{N}_K(X_q) = \arg \min_{S \subset \mathcal{D}, |S|=K} \sum_{d \in S} \|\text{vec}(X_q) - \text{vec}(X_d)\|_2. \quad (3)$$

For PEMS07M, $\text{vec}(X) \in \mathbb{R}^{196 \times 608}$ ($= 288 \text{ timesteps} \times 228 \text{ nodes} \times 3 \text{ features}$). This simple approach improves sharply over random (MAE 6.14 $\rightarrow$ 4.66 for $K = 3$), yet still does not surpass zero-shot. Raw L2 treats every element as an independent coordinate, so proximity reflects point-wise numerical closeness rather than shared traffic state such as congestion regime or rush-hour pattern. The retrieved residuals c_k are therefore often uninformative, limiting the corrector's signal-to-noise ratio.

Embedding-space KNN: We retrieve nearest neighbors in the frozen encoder's feature space after mean-pooling over nodes. Let $h_n \in \mathbb{R}^{512}$ be the per-node feature produced by the encoder for the window. We pool across nodes and retrieve in the pooled space,

$$\bar{h} = \frac{1}{N} \sum_{n=1}^N h_n \in \mathbb{R}^{512}, \quad (4)$$

$$\mathcal{N}_K(X_q) = \arg \min_S \sum_{d \in S} \|\bar{h}_q - \bar{h}_d\|_2. \quad (5)$$

The encoder features come from the same Phase-1 cache used for corrections. Proximity in this space reflects similarity in learned traffic representations, capturing shared congestion regimes, rush-hour patterns, and spatial dependencies rather than point-wise numerical closeness. This design is inspired by kNN-LM [8], which demonstrated that retrieving nearest neighbors in a pretrained model's representation space at inference time improves predictions without any parameter updates — an analogous principle to our embedding-space correction strategy. We mean-pool to a single 512-d vector per window to keep retrieval cheap. Under the same learned aggregator, swapping raw-L2 for embedding KNN reduces PEMS07M MAE from 4.30 to 4.15, an extra 3.5% relative gain from the retrieval step alone (Fig. 2).

Scope of embedding KNN in this paper. We use embedding KNN as the in-distribution upper bound on PEMS07M, but keep raw-L2 KNN for SZ-DIDI and CD-DIDI. The core contribution of this paper is the residual correction framework itself, not the retrieval metric. Raw-L2 already demonstrates the framework's effectiveness on OOD cities. Under severe out-of-distribution shifts (such as shifting from California highway speeds to Shenzhen ride-hailing demand), the representation space of the frozen encoder may exhibit domain drift or geometric anisotropy. If the latent features undergo severe representation degradation, embedding-space retrieval is prone to selecting semantically misaligned demonstrations, introducing out-of-distribution noise into the aggregator, and leading to performance degradation. Our future study further explores this representation-retrieval trade-off and develops enhancements that improve the generalizability of latent retrieval under extreme modality shifts.

3.4 Cross-Attention Aggregator

Naive uniform averaging assigns equal weight $1/K$ to each demonstration's residual. This dilutes informative corrections with noise from less-relevant demonstrations and, as our results confirm, fails to surpass zero-shot performance on its own. We replace it with a learned aggregation module. We compare a minimal 8K-parameter cosine-similarity variant (SIMPLEDEMOAGGREGATOR) against a 198K-parameter cross-attention variant (DEMOAGGREGATOR); the latter is our main design, detailed below and illustrated in Fig. 3.

Inputs. The aggregator receives three tensors drawn from the cached Phase-1 outputs:

$$\begin{aligned} h_q &\in \mathbb{R}^{B \times N \times D}, & (\text{query features, } D = 512) \\ \{h_k\} &\in \mathbb{R}^{B \times K \times N \times D}, & (\text{demo features}) \\ \{c_k\} &\in \mathbb{R}^{B \times K \times T' \times N \times 1}. & (\text{demo residuals}) \end{aligned}$$

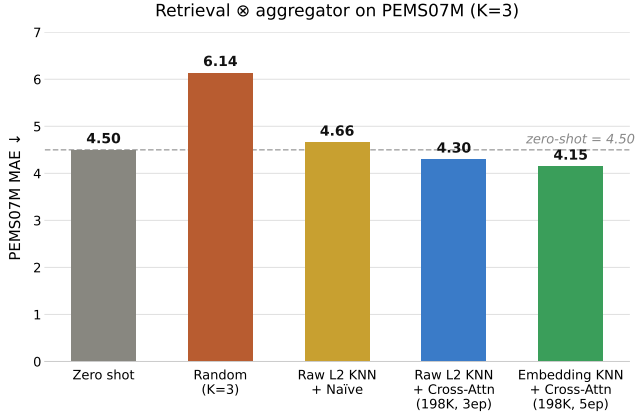


Figure 2: PEMS07M MAE across retrieval@aggregator combinations (K=3): zero-shot, random, raw-L2 KNN with naive averaging, raw-L2 KNN with cross-attention, and embedding KNN with cross-attention.

Per-node, per-head attention over demonstrations. Queries Q and keys K are obtained by layer-normalising the encoder features and projecting them to $H \cdot d_h$ dimensions ($H = 4$ heads, $d_h = 32$, internal $d_{\text{proj}} = 128$):

$$Q = W_Q \text{LN}(\mathbf{h}_q) \in \mathbb{R}^{B \times N \times H \times d_h}, \quad (6)$$

$$K = W_K \text{LN}(\mathbf{h}_k) \in \mathbb{R}^{B \times K \times N \times H \times d_h}. \quad (7)$$

For every batch element b , node n , and head h , we softmax over the K demonstrations:

$$\alpha_{b,n,h,k} = \text{softmax}_k \left(\frac{Q_{b,n,h} \cdot K_{b,k,n,h}^\top}{\sqrt{d_h}} \right). \quad (8)$$

This yields a *per-node* demo-weight vector per head, which is critical for spatio-temporal data where different sensors may match different demonstrations.

Values are the raw corrections (no V-projection). Standard cross-attention projects values through a learned W_V before weighting. Here, the corrections $\{c_k\}$ already live in the prediction space $\mathbb{R}^{T' \times N \times 1}$, so we use them directly as values with no projection. The attention weights α from Q/K determine how much each demo's correction contributes, and the weighted sum produces a candidate correction per head:

$$\tilde{\mathbf{c}}_{b,t,n}^{(h)} = \sum_{k=1}^K \alpha_{b,n,h,k} \mathbf{c}_{b,k,t,n} \in \mathbb{R}^{B \times T' \times N}. \quad (9)$$

Head combination. A linear layer with no bias, initialised to $1/H$, merges the H per-head candidates,

$$\tilde{\mathbf{c}}_{b,t,n} = \sum_{h=1}^H w_h \tilde{\mathbf{c}}_{b,t,n}^{(h)} \quad \text{where } w_h \in \mathbb{R}. \quad (10)$$

The init-to-average lets the aggregator recover uniform head-averaging at step 0, then deviate from it as training progresses.

Query-conditioned output gate. A per-node multiplicative gate $g_n \in \mathbb{R}_{\geq 0}$ is computed from the query feature by a small two-layer MLP with softplus output and *zero-initialised* final layer:

$$g_n = \text{softplus}(W_2 \text{SiLU}(W_1 \mathbf{h}_{q,n})), \quad W_2 \leftarrow \mathbf{0}. \quad (11)$$

The hidden activation is SiLU ($x \cdot \sigma(x)$), also known as Swish). Zero-initializing the final layer ensures that $g_n \approx \text{softplus}(0) \approx 0.693$ at the start of training, so the aggregator initially applies a mild, uniform scaling to the correction rather than collapsing to zero or saturating. This stable initialization allows the gate to gradually learn when to amplify or suppress corrections as training progresses. Softplus is chosen over sigmoid to allow *amplification* of corrections (gate > 1), which is empirically useful under large distribution shifts (Section 4).

Output. The final per-query correction is

$$\Delta_q = g \cdot \tilde{\mathbf{c}} \in \mathbb{R}^{B \times T' \times N \times 1}, \quad \hat{Y}_q = f_\theta(\mathbf{X}_q) + \Delta_q. \quad (12)$$

Parameter budget. LN: $2D = 1,024$; W_Q : $D \cdot 128 + 128 = 65,664$; W_K : $65,664$; head-combine: 4 ; gate-MLP ($D \rightarrow 128 \rightarrow 1$, zero-init): $65,793$. Total: $\approx 198K$, i.e. $\approx 0.8\%$ of the 26M-parameter OpenCity-plus backbone.

3.5 Training Procedure

Training proceeds in two phases. *Phase 1* (one-time) runs the frozen base model forward over all training windows and caches, for every $d \in \mathcal{D}$, the prediction $f_\theta(\mathbf{X}_d)$, the encoder features $\mathbf{h}_d$ (both pooled and non-pooled), and the correction $\mathbf{c}_d$. *Phase 2* trains the aggregator from cached data only, with no forward or backward pass through the base model. We use Adam at learning rate 10^{-4} , MAE loss, and early stopping with patience 5. Phase 1 is amortised across all downstream experiments (different K , retrievers, and aggregators), making the ablations in Section 4 tractable.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on three traffic datasets spanning two distribution regimes. *PEMS07M* contains 228 highway speed sensors in California at 5-minute intervals (50/10/40 train/val/test split, 5,761 training windows). *PEMS07M* is not included in OpenCity's pre-training corpus, but shares the same modality of highway speed as several pretraining datasets. Notably, OpenCity itself uses *PEMS07M* as a zero-shot evaluation target [9], meaning the base model has never seen this specific sensor network during pretraining despite being trained on similar highway speed data. We therefore treat it as our *in-distribution* benchmark, where the backbone has seen similar data but not this specific sensor network. *SZ-DIDI* (Shenzhen, 627 nodes) and *CD-DIDI* (Chengdu, 524 nodes) are ride-hailing demand datasets at 10-minute intervals, used by OpenCity's paper to report fast-adaptation results. These datasets represent a challenging *out-of-distribution* regime along three axes: unseen cities (Shenzhen and Chengdu vs. California), an unseen traffic modality (ride-hailing demand versus highway speed), and a coarser temporal resolution (10-minute vs. 5-minute intervals).

Base model. OpenCity-plus [9] with embedding dimension 512, encoder depth 6, 16 attention heads, and $\sim 26M$ parameters. For *DIDI*

Cross-attention aggregator

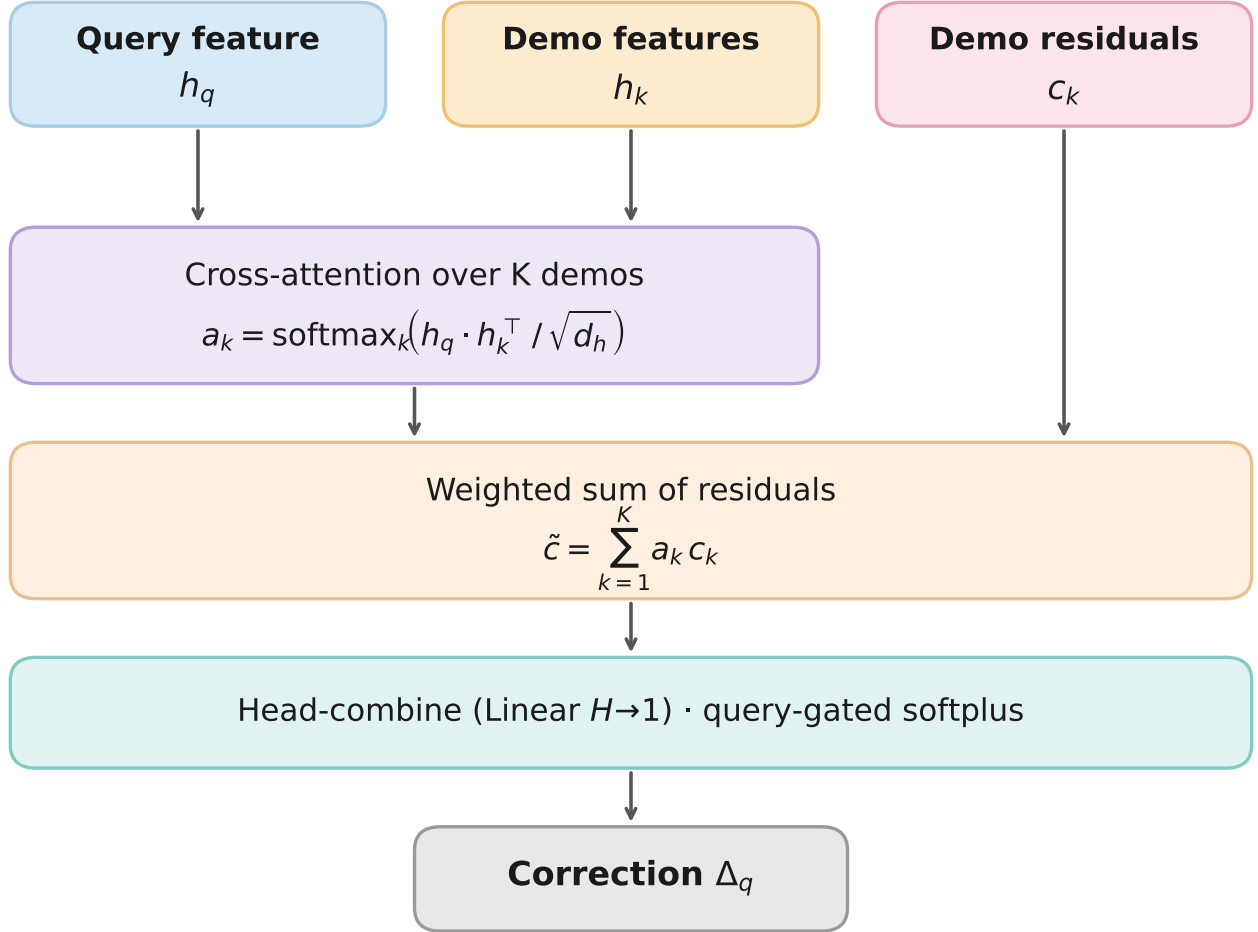


Figure 3: Cross-attention aggregator: softmax attention between query and demo features (Q, K) weights cached demo residuals (V), then a head-combine with a query-gated softplus produces the correction Δ_q .

datasets we use input/output window = 144 steps ($288/(10/5)$) with PatchEmbedding gap = 2, as prescribed by the OpenCity recipe.

Metrics. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE, %), and Pearson correlation (CORR), computed with the standard masked evaluation (mae threshold = 0, mape threshold = 0.001).

Implementation. PyTorch 2.4.1 on NVIDIA L40S (46 GB), fp32 precision throughout.

4.2 Main Results on PEMS07M

Table 1 summarises our main PEMS07M results, from which four observations stand out. First, random demonstrations significantly degrade performance (MAE 4.50 $\rightarrow$ 6.14, -36% relative to zero-shot),

confirming that uncorrelated residuals add noise rather than signal. Second, raw-input KNN substantially reduces this degradation (MAE 6.14 $\rightarrow$ 4.66) but still does not surpass zero-shot, demonstrating that structured retrieval is necessary but not sufficient on its own. Third, learned aggregation surpasses zero-shot, with SIMPLEDEMOAGGREGATOR reaching 4.47 and DEMOAGGREGATOR reaching 4.30. Fourth, swapping raw-L2 for embedding-space KNN under the same learned aggregator delivers the best result (MAE 4.15, RMSE 7.61, MAPE 11.25%, CORR 0.739), a 7.8% MAE gain over zero-shot with only 0.8% extra parameters and no modification to the backbone.

Table 1: Main results on PEMS07M. All ICL methods use $K=3$. Best in bold. Dashes for full-shot baselines indicate metrics not reported in [9].

Method	Retrieval	Params	MAE↓	RMSE↓	MAPE%↓	CORR↑
Zero-Shot	—	0	4.50	8.21	12.20	0.736
ICL-Traffic / Random	random	0	6.14	9.53	14.84	0.634
ICL-Traffic / Naive avg	raw-L2	0	4.66	8.11	11.92	0.729
ICL-Traffic / Simple agg	raw-L2	8K	4.47	8.01	11.55	0.737
ICL-Traffic / Cross-attn (3ep)	raw-L2	198K	4.30	7.68	11.37	0.748
ICL-Traffic / Cross-attn (3ep)	emb-KNN	198K	4.15	7.61	11.25	0.739
<i>Full-shot supervised baselines (as reported by [9])</i>						
GWN [18]	—	full	4.17	—	—	—
ASTGCN [7]	—	full	4.39	—	—	—
STGCN [19]	—	full	4.44	—	—	—

Table 2: Cross-dataset adaptation. ICL aggregator and fine-tune both use 3 epochs on target-domain training data. Best in bold, best frozen-backbone in underline.

Dataset	Method	Backbone	MAE↓	RMSE↓	MAPE%↓	CORR↑
SZ-DIDI	Zero-Shot	frozen	4.66	7.10	19.05	0.322
	ICL-Traffic	frozen	<u>2.70</u>	<u>4.11</u>	<u>11.08</u>	<u>0.540</u>
	Fine-tune head (3ep)	modified	2.33	3.75	9.89	0.681
CD-DIDI	Zero-Shot	frozen	6.34	9.36	28.36	0.336
	ICL-Traffic	frozen	<u>3.41</u>	<u>5.02</u>	<u>15.28</u>	<u>0.720</u>
	Fine-tune head (3ep)	modified	2.60	3.93	11.98	0.793

4.3 Cross-Dataset Generalization: ICL vs. Fine-Tuning

A more important question is whether output-space ICL can adapt to genuinely out-of-distribution cities and how it compares to gradient-based fine-tuning at the same budget. Table 2 and Fig. 4 report this comparison on SZ-DIDI and CD-DIDI. The two adaptation methods (our 3-epoch cross-attention aggregator and OpenCity’s 3-epoch prediction-head fine-tuning) use the same target-domain training set and the same 3-epoch budget, with zero-shot as the no-adaptation reference. The only difference between the two is how the training data is used.

A notable pattern emerges: ICL-Traffic’s relative improvement over zero-shot scales with the degree of distribution shift. On in-distribution PEMS07M the aggregator reduces MAE by 4.4–7.8%, whereas on the strongly out-of-distribution SZ-DIDI and CD-DIDI datasets the gains reach 42% and 46% respectively. Under strong distribution shift, the backbone exhibits systematic, consistent errors such as under-prediction of demand peaks that are shared across similar inputs. KNN-retrieved demonstrations expose precisely these consistent biases, giving the aggregator a high-signal correction that generalizes across the test set.

We quantify how much of fine-tuning’s gain is reachable from output space alone. Define

$$\text{Recovery} = \frac{\text{MAE}_{\text{ZS}} - \text{MAE}_{\text{ICL}}}{\text{MAE}_{\text{ZS}} - \text{MAE}_{\text{FT}}}. \quad (13)$$

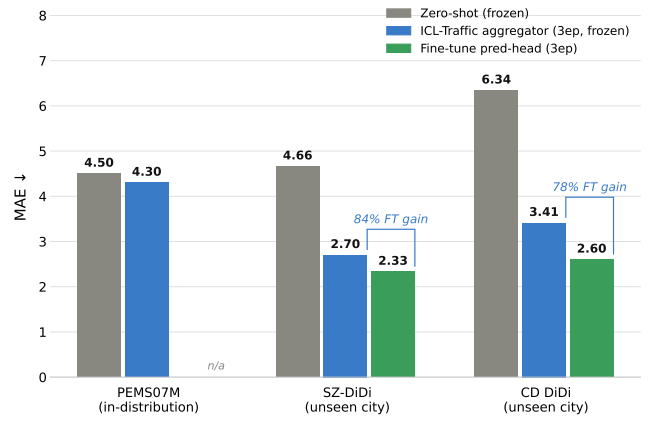


Figure 4: Cross-dataset adaptation on SZ-DIDI and CD-DIDI (3-epoch budget): zero-shot, frozen-backbone ICL-Traffic aggregator, and prediction-head fine-tuning. FT-gain recovery is annotated above each aggregator bar.

Table 3: Ablation: demonstration selection strategy. All with naive averaging. S : sampling rounds.

Selection	K	S	MAE	RMSE	vs. ZS
Random	1	1	7.39	12.00	−64%
Random	3	1	6.14	9.53	−36%
Random	1	10	5.24	8.52	−16%
KNN (raw-L2)	1	1	5.61	10.04	−25%
KNN (raw-L2)	3	1	4.66	8.11	−3.6%

ICL recovers 84% on SZ-DIDI and 78% on CD-DIDI with a fully frozen backbone. The remaining ~20% gap is structural: it corresponds precisely to the benefit of remapping encoder representations to a new output distribution — a transformation that no output-space method can achieve by construction. To our knowledge, ICL-Traffic provides the first empirical decomposition of adaptation gain into output-space-recoverable and representation-dependent components for any ST-FM.

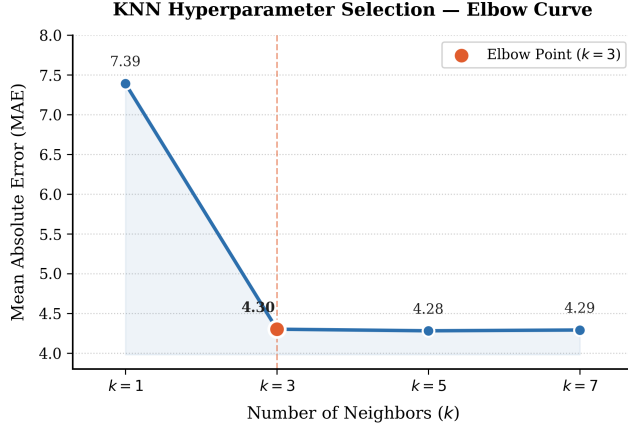
4.4 Ablation: Demonstration Selection Strategy

Table 3 isolates the effect of demonstration retrieval with naive averaging. Random selection with $K=1$ yields MAE 7.39 (−64% vs. zero-shot). Increasing to $K=3$ reduces variance but remains far below zero-shot (−36%). Multi-round sampling ($S=10$) with random selection partially recovers by averaging out noise (−16%) but cannot match KNN. With KNN, even $K=1$ (MAE 5.61) outperforms random $K=3$, and $K=3$ (MAE 4.66) approaches zero-shot. Demo-query alignment is the dominant factor.

Fig. 5 visualizes the effect of k on MAE under the cross-attention aggregator with raw-L2 KNN retrieval (Table 4). At $k=1$, MAE spikes to 7.39, well below zero-shot, because a single demonstration carries high residual variance and no averaging effect. The sharpest gain occurs between $k=1$ and $k=3$ (MAE 7.39 → 4.30), after which the curve plateaus: $k=5$ and $k=7$ yield MAE 4.28 and

Table 4: Ablation: demonstration selection strategy. All with cross attention. Selection : KNN (raw-L2)

K	MAE	RMSE	vs. ZS
1	7.39	12.00	−64.2%
3	4.30	7.68	+4.4%
5	4.28	7.69	+4.8%
7	4.29	7.74	+4.6%

**Figure 5: Elbow curve illustrating the effect of the number of nearest neighbors (k) on Mean Absolute Error (MAE) for the KNN (raw L2) approach. MAE drops sharply from $k = 1$ (7.39) to $k = 3$ (4.30), after which it plateaus, indicating $k = 3$ as the optimal hyperparameter.**

4.29 respectively, a difference of less than 0.01. This elbow pattern reflects a signal-to-noise trade-off: additional neighbors beyond $k=3$ contribute diminishing correction signal while introducing slightly less-similar demonstrations whose residuals add marginal noise. We therefore fix $k=3$ as the default across all experiments, balancing correction quality against retrieval overhead.

4.5 Ablation: Aggregation Architecture

Table 5 compares aggregation strategies at $K=3$. Naive averaging (0 parameters) achieves MAE 4.66 but cannot surpass zero-shot. SIMPLEDEMOAGGREGATOR (8K) learns cosine-similarity weights and surpasses zero-shot (4.47). DEMOAGGREGATOR (198K) captures richer query–demo interactions and reaches 4.30 at 3 epochs; continuing to 10 epochs adds only 0.01 MAE, confirming rapid convergence from cached features. Replacing raw-L2 with embedding-KNN under the same 198K aggregator delivers the overall best result of 4.15 in 5 epochs.

4.6 Analysis

Why random demonstrations hurt. Random corrections are uncorrelated with the query’s prediction error, so averaging them just adds noise. As $S \rightarrow \infty$, the average correction converges to the population mean residual, which Phase 1 statistics confirm is

Table 5: Ablation: aggregation architecture on PEMS07M. All with $K=3$.

Aggregator	Retrieval	Params	MAE↓	RMSE↓	MAPE%↓	CORR↑
Naive avg	raw-L2	0	4.66	8.11	11.92	0.729
Cosine (5ep)	raw-L2	8K	4.47	8.01	11.55	0.737
Cross-attn (3ep)	raw-L2	198K	4.30	7.68	11.37	0.748
Cross-attn (10ep)	raw-L2	198K	4.29	7.71	11.42	0.748
Cross-attn (3ep)	emb-KNN	198K	4.15	7.61	11.25	0.739

nearly zero, so random-demo ICL recovers zero-shot at best and never surpasses it.

Why KNN is necessary but not sufficient. Similar inputs yield correlated prediction errors, so KNN-retrieved corrections carry real signal. Naive uniform averaging, however, dilutes this signal because even among the K nearest neighbors, residual noise remains, and nodes that match the query well average with nodes that match poorly. A learned weighting is needed to focus on the most relevant corrections and to do so *per node*.

Why learned aggregation surpasses zero-shot. The aggregator learns per-node, per-head weights that up-weight the most informative demonstrations. Even the 8K-parameter simple aggregator is enough to beat zero-shot, confirming that the missing ingredient on top of KNN is weighting rather than any richer transformation. The 198K cross-attention variant then captures richer interactions through its multi-head structure and query-conditioned gate, yielding the largest further gain.

Convergence. The aggregator converges by epoch 3–4 from cached features. The 3-epoch row (MAE 4.30 on PEMS07M) is deliberately aligned with OpenCity’s 3-epoch fast-adaptation protocol so the cross-dataset comparison (Section 4.3) is budget-matched; extending to 5 epochs only matters when retrieval quality has improved (embedding KNN).

Failed approach: naive sequence extension. We initially attempted standard ICL by concatenating demonstrations into the model’s input sequence. This caused out-of-distribution self-attention patterns and positional encoding mismatches, degrading MAE by 9–28% versus zero-shot. This failure motivated our output-space approach.

Per-query retrieval vs. fine-tuning. Fine-tuning treats all training examples equally and produces a single globally adapted model. Our method selects the few most similar examples per query and weights their corrections accordingly, preserving local structure that a global update would average out. This explains why ICL’s relative gain over zero-shot grows with the size of the distribution shift.

5 Discussion and Future Work

5.1 Limitations

Compute overhead. Our method nominally requires $K+1$ forward passes per test sample ($4\times$ for $K=3$). In deployment, all demonstration passes are precomputed once in Phase 1. The marginal test-time cost is then the single query pass plus a KNN lookup plus a 198K-parameter aggregator call, which is negligible.

Relationship to true in-context learning. In ICL-Traffic, demonstrations are processed independently through the frozen backbone rather than jointly with the query in a shared context window. This design is not merely a compromise; it is a direct response to the fixed-length positional encoding of current ST-FMs, which causes severe out-of-distribution attention patterns when sequences are extended (Section 4.6). As ST-FM architectures evolve to support variable-length or cross-sequence attention, true input-space ICL becomes a natural extension of our framework. True ICL, where the model attends jointly over demonstrations and query as in ICT [3] and TimesFM-ICF [4], would require architectural modification and continued pretraining.

Contrast to TTA methods. Unlike parameter-updating TTA methods (e.g., TENT [17] and TTT [16]) that require online backward passes and weight modifications, ICL-Traffic keeps its backbone frozen with zero test-time gradient computation. This distinction is critical for three reasons. First, standard TTA incurs substantial computational and memory overhead, making it impractical for resource-constrained edge devices. Second, TTA’s unsupervised signals (e.g., entropy minimization) degrade under non-stationary streams or severe structural domain shifts (e.g., transition from California highway speeds to Shenzhen ride-hailing demand) where the baseline output distribution is fundamentally misspecified; ICL-Traffic sidesteps this by using labeled in-context demonstrations to directly estimate and correct systematic bias. Lastly, we view ICL-Traffic and TTA as orthogonal paradigms [3]: output-space ICL eliminates coarse-grained spatial and modal biases at zero gradient cost. In the future, we would like to pursue a promising hybrid pipeline that applies ICL-Traffic first to correct the bulk of the shift, followed by lightweight TTA on the residual error.

Retrieval ceiling. Raw-input L2 matches by point-wise numerical closeness, not by traffic-pattern similarity. Embedding-space KNN is better because the encoder captures task-relevant structure, but the encoder was not trained for retrieval, so nearby demos in feature space are not necessarily the ones whose corrections help most. We discuss directions for closing this gap in Section 5.2.

Trainable parameters. Our aggregator requires an offline training phase from cached features. Like fine-tuning, we use target-domain training data. The difference is in *how* that data is used: fine-tuning absorbs it into model weights; ICL-Traffic preserves it as a retrievable demo pool and the aggregator learns only a general weighting strategy. At inference, per-query variation comes from demo retrieval, not from aggregator weights (which are fixed).

Capability ceiling. Output-space correction recovers $\sim 80\%$ of head fine-tuning’s MAE gain on strongly OOD cities. The remaining 20% requires updating the prediction head, which output-space

methods cannot reach by construction — the encoder’s internal representations remain mapped to the pretraining output distribution regardless of how accurately the aggregator corrects individual predictions. This gap has a precise interpretation: it represents the mutual information between the encoder’s representation and the target city’s output distribution that is not captured in prediction space alone. A natural next step is a two-stage approach: apply ICL-Traffic first to recover the output-space-reachable gain, then fine-tune only the prediction head (not the encoder) on the residual error that ICL leaves behind. This would require fewer gradient steps than full fine-tuning since ICL-Traffic has already corrected the bulk of the distribution mismatch.

5.2 Future Work

Better retrieval. Several directions can narrow the retrieval ceiling, including training a task-aware retrieval metric that optimises for correction usefulness, incorporating temporal priors, and exploring distance functions beyond L2. Concurrent retrieval-augmented approaches such as RAST [14] validate the broader premise of retrieval-based spatio-temporal forecasting and suggest promising directions for combining our output-space correction with dual-dimension memory banks that index spatial and temporal pattern dimensions independently.

Richer aggregation. The aggregator could learn horizon- or graph-aware weights, for example a GCN layer that smooths per-node attention over the road network.

5.3 Practical Deployment Considerations

ICL-Traffic is designed with real-world traffic management deployment in mind. Three properties make it particularly suited to operational intelligent transportation systems.

Gradient-free inference. The frozen backbone requires no infrastructure for gradient computation at deployment time; the aggregator and demonstration pool can be served from a standard inference endpoint without GPU-resident optimizer state, reducing deployment cost and complexity compared to fine-tuning-based adaptation.

Updatable demonstration pool. The demonstration pool is naturally updatable: as new traffic observations accumulate in a city, the pool can be extended without retraining, allowing the system to adapt to seasonal traffic pattern shifts (e.g., school year vs. summer, or special events) by simply refreshing the pool with recent labeled windows.

Interpretable corrections. The per-query retrieval mechanism provides a natural explanation for each correction; the K retrieved demonstrations can be surfaced to traffic operators as “similar historical situations,” grounding the model’s adaptation in interpretable precedents rather than opaque weight updates. This transparency is particularly valuable in safety-critical applications such as adaptive signal control, where operators need to understand and trust model behavior.

These properties collectively suggest that ICL-Traffic is well-suited for deployment in adaptive traffic signal control systems, where models must generalize across intersections with heterogeneous sensor coverage and traffic characteristics without per-intersection retraining.

6 Conclusion

We have presented a lightweight, gradient-free in-context learning application for traffic prediction (ICL-Traffic) using pretrained spatio-temporal foundation models that keep the underlying backbone completely frozen. Our framework is motivated by the failure of traditional input-space ICL in this domain, where concatenating demonstrations degrades performance by 9% to 28% due to out-of-distribution attention patterns and sequence length mismatches. Our experiments demonstrate that demonstration retrieval and learned output-space aggregation are both necessary and jointly sufficient to act as a highly effective, gradient-free adaptation surrogate. Importantly, we reveal that the benefits of this output-space correction scale with the severity of the distribution shift, recovering 78% to 84% of the MAE gains of gradient-based prediction-head fine-tuning on out-of-distribution cities by isolating and correcting structured systematic biases. This is achieved with only 0.8% additional parameters and zero test-time gradient updates. The remaining $\sim 20\%$ gap defines a structural ‘Representation Adaptation Gap,’ representing the boundary where post-hoc output corrections must yield to representation-level parameter alignment. This limitation motivates future work exploring hybrid two-stage pipelines and methods to stabilize embedding-space retrieval against geometric anisotropy under severe domain shifts. We release our code and model weights to support reproducibility and encourage extension to other spatio-temporal foundation architectures.

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